COSC2779/COSC2972 Deep Learning:
Exam Example Questions

This document provides sample questions for the three question types that may appear in the COSC2779/COSC2972 Deep Learning Exam.

These examples are intended to help you understand the style and format of the exam questions. The actual exam will contain a different number of questions and different question content.

Students should use these examples as a guide only and ensure they are familiar with all course materials and learning outcomes assessed in the course.

PART A – MULTIPLE CHOICE QUESTIONS

The following multiple-choice questions may have multiple correct answers. You must choose ALL the correct options while NOT selecting any of the incorrect options. No explanation is required.

For each question, the penalty for selecting an incorrect option is equal to the mark awarded for a correctly selected option. The score for a question cannot be less than 0.

Question A.1:

Which of the following are properties of convolutional neural networks?

(i) Sparse connectivity (feature extraction usually happens locally)
(ii) Parameter sharing (the same operation is applied at different locations)
(iii) Dense connectivity (every output is connected to all input locations)
(iv) Pooling helps reduce redundant information and provides some level of invariance to translations

Solution: (i), (ii), (iv)

Question A.2:

During fine-tuning in transfer learning, which of the following are correct steps?

(i) Unfreeze some top layers of the base model
(ii) Train the unfrozen base model layers with a smaller learning rate
(iii) Both the head and base model weights can be trained
(iv) All layers in the base model must remain frozen

Solution: (i), (ii), (iii)

Question A.3:

You are given the task of predicting the genre of short music clips of different lengths. Which Recurrent Neural Network (RNN) architecture is most suitable for this task?

(i) One-to-one
(ii) One-to-many
(iii) Many-to-one
(iv) Many-to-many

Solution: (iii)

Question A.4:

Which of the following are issues with a basic autoencoder?

(i) It can memorize (over-fit) the training data
(ii) It cannot reliably generate novel samples (e.g. images) because its latent space distribution is unconstrained
(iii) It cannot reconstruct its input
(iv) It requires adversarial training

Solution: (i), (ii)

Question A.5:

Which of the following statements about activation functions are correct?

(i) Sigmoid outputs are always strictly between 0 and 1.
(ii) ReLU outputs are always greater than or equal to 0.
(iii) Sigmoid has a large gradient when its inputs have large positive or large negative values.
(iv) ReLU has a large gradient when its inputs have large negative values.

Solution: (i), (ii)

Question A.6:

Which of the following statements about GRUs and LSTMs are correct?

(i) Both have built-in attention mechanisms.
(ii) Both are designed to help capture long-range dependencies.
(iii) Neither has a cell state.
(iv) Both can process sequential data.

Solution: (ii), (iv)

Question A.7:

Which of the following statements about GANs are correct?

(i) Training GANs involves training both a generator and a discriminator.
(ii) GANs are notoriously difficult to train.
(iii) A trained GAN requires both the generator and discriminator to generate new images.
(iv) Convolutional neural networks can be used in both the generator and discriminator of GANs.

Solution: (i), (ii), (iv)

Question A.8:

Which of the following statements about diffusion models are correct?

(i) Diffusion models generate new samples (e.g. images) through multiple denoising steps.
(ii) Diffusion models are generally unstable to train.
(iii) Generating new samples (e.g. images) using diffusion models is generally fast.
(iv) Diffusion models generally generate more diverse images compared to GANs.

Solution: (i), (iv)

PART B – SHORT ANSWER QUESTIONS

The questions in this section can be answered in 2–4 sentences. Please keep your responses concise and focused.

Question B.1:

You are designing a neural network for image classification. A colleague suggests replacing all convolutional layers with fully connected layers. Why might convolutional layers be more suitable for image data?

Sample Solution:

Convolutional layers use sparse connectivity and parameter sharing. They extract features locally and apply the same operation at different locations. This exploits the spatial structure of image data, thus requiring fewer parameters than fully connected layers, and allowing CNNs to learn image features more effectively and potentially achieve better performance.

Question B.2:

In a recurrent encoder–decoder model without attention, the decoder relies on the final encoder state to represent the entire input sequence. What limitation can this cause, particularly for long sequences, and how does attention help to address this limitation?

Sample Solution:

The final encoder state may not capture all important information from the input sequence, particularly for long sequences. Attention allows the decoder to look back at all encoder states (not just the last one). This gives the decoder access to information from different parts of the input sequence, helping it focus on the most relevant parts of the input when generating each output.

Question B.3:

You have a pre-trained Transformer and a small labeled dataset for a new classification task. You want to adapt the model while training as few parameters as possible. Would you use feature extraction or full fine-tuning? Briefly explain why and how you would do this.

Sample Solution:

Feature extraction is more suitable. We freeze the backbone (pre-trained Transformer model), use its outputs as fixed features, and train only a small task-specific head for the classification task. This requires training fewer parameters, is faster than full fine-tuning, and is suitable for small datasets.

Question B.4:

You train a feedforward neural network using stochastic gradient descent (SGD), but the training loss oscillates significantly and does not seem to decrease and converge. What might cause this problem and how could you address it?

Sample Solution:

One possible cause is that the learning rate is set too high, causing the model's parameters to overshoot good solutions during optimization. To address this, we can reduce the learning rate or use a learning rate schedule that gradually decreases it during training.

Question B.5:

You train a basic autoencoder on a dataset of images. After training, you want to generate a new image by randomly sampling a latent code z and passing it through the decoder. What problem might you encounter and how could you overcome it?

Sample Solution:

For a basic autoencoder, the distribution of latent codes is not explicitly constrained, so randomly sampling a latent code may produce an unrealistic image. To address this, we can use a Variational Autoencoder (VAE) which encourages the latent codes to follow a known prior distribution (e.g. a standard normal distribution). This allows us to sample latent codes from that distribution and pass them through the decoder to generate more realistic images.

Question B.6:

You need to adapt the same large foundation model to multiple downstream tasks, but storing a separate fully fine-tuned model for each task would be too expensive. What approach would you use and why?

Sample Solution:

I would use a Parameter-Efficient Fine-Tuning (PEFT) technique such as LoRA (Low-Rank Adaptation). LoRA keeps the pre-trained foundation model's weights frozen and trains only a small number of parameters through low-rank updates (LoRA parameters) for each task. This reduces the number of trainable parameters and allows us to store only one copy of the foundation model, together with the task-specific LoRA parameters for each task.

PART C - PROBLEM-SOLVING QUESTIONS

The questions in this section involve multi-step problems based on practical deep learning scenarios. Answer each part clearly and show your reasoning where appropriate.

Problem C.1 - Developing a CNN Image Classifier

You are developing a deep learning system to classify images of recyclable waste into six categories. You have several thousand labelled RGB images and decide to develop a CNN-based classifier.

(a) Choosing a CNN

A team member suggests using a fully connected neural network directly on the image pixels instead of a CNN. Why is a CNN more suitable for this task? Give two reasons.

Sample Solution:

CNNs are more suitable for images because they have several important properties:

- Sparse connectivity: feature extraction happens locally.
- Parameter sharing: the same operation is applied at different locations.

These properties exploit the spatial structure of image data, thus requiring fewer parameters than fully connected layers, and allowing CNNs to learn image features more effectively and potentially achieve better performance.

(b) Initial training

You split your dataset into training, validation, and test sets. For your first attempt, you train a CNN from scratch and obtain the following results:

- Training accuracy: 72%
- Validation accuracy: 70%

Both training and validation performance are lower than you expected. Does this result mainly suggest overfitting or underfitting? Give one change you could try to address this problem.

Sample Solution:

This result mainly suggests underfitting, because the model does not perform well even on the training set, and there is only a small difference between training and validation performance. One possible approach is to increase the model capacity, for example by adding more layers or filters.

(c) Improving generalization

After increasing the capacity of the CNN and training it again, you obtain:

- Training accuracy: 96%
- Validation accuracy: 78%

What problem does this result suggest? Briefly explain why and give two techniques that could help address this problem.

Sample Solution:

This result suggests overfitting, because the model performs very well on the training set but there is a large gap between the training and validation performance. Possible techniques that can address this issue include data augmentation, dropout, and regularization.

(d) Evaluating the classification performance

After improving the model, you obtain good overall validation accuracy. However, you want to understand whether the CNN performs well across all six waste categories, and which categories it tends to confuse. What additional evaluation results could you examine, and what information would they provide?

Sample Solution:

We can examine per-class performance (e.g., accuracy, precision, recall) and a confusion matrix. Per-class performance shows us whether the model performs well or poorly on each individual waste category. A confusion matrix compares the ground-truth and predicted categories, allowing us to identify which categories are commonly confused with each other.

(e) Using a pretrained model

You later obtain access to an ImageNet-pretrained ResNet. Instead of continuing to train your CNN from scratch, describe how you could adapt the pretrained ResNet to this classification task.

Sample Solution:

We can use transfer learning. We replace the original classification head of the pretrained ResNet with a new classification head for the six waste categories. Initially, we freeze the pretrained layers and train only the new classification head using our labelled data. Then, if we want to further improve the performance, we can unfreeze some of the pretrained layers of the pretrained ResNet and continue to fine-tune them on our labelled dataset using a smaller learning rate.